

Graph-Structured Lambda Theories

A Reference for the Working Developer

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Abstract

A graph-structured lambda theory (GSLT) is a language presentation: a signature of sorts and operators, a set of equations, and a set of rewrite rules. This document is a reference account of what can be built once a language is presented that way. We develop the definition and two successive enrichments — *interactive* GSLTs, which name a site of interaction, and *continued* interactive GSLTs, whose interaction is presented as a meterable cut — and we discriminate the two with worked examples, including examples that fail. We then assemble the category **GSLT**, whose morphisms preserve bisimulation over context-labelled transitions, and exhibit three constructions on it: the cost monad $\mathcal{C}$, which meters a theory; the history monad $\mathcal{H}$, which logs it; and the Hypercube functor, a member of the OSLF family of algorithms, which generates a family of type systems from the theory’s own rewrite rules. Finally we show that all of this is programmable. The **language!** block of Rholang 1.4 is a data definition language whose data can wiggle: without equations or rewrites it presents algebraic data types, with equations it presents finitely presentable universal algebra, and with rewrites it presents domain-specific languages. The control flow language is a Rholang whose channels are typed by such specifications, and whose **for**-comprehension carries a **where** clause guarding communication on context-labelled modal formulae drawn from the generated logic.

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1 Introduction

1.1 The claim

A language is a triple. Sorts and operators say what can be written; equations say which writings are the same; rewrites say how writings move. Everything this document develops — a type system, a cost discipline, a history log, a runtime, a logic for stating properties of programs — is generated from that triple, and generated uniformly, by constructions that do not know which language they are applied to.

That is an ambitious claim and it is worth saying immediately what it does not mean. It does not mean that every interesting question about a language is answered by its presentation; it means that a surprising number of the artifacts we are used to building by hand, once per language, are functorial in the presentation. The cost of adding cost accounting to a new calculus should not be a research programme. The cost of getting a type system for a new domain-specific language should not be a semester. If the presentation is the input, these become instances of one construction rather than a series of fresh inventions.

1.2 Who this is for

Two audiences, addressed simultaneously and, we hope, without either being short-changed.

The first is the reader who wants the mathematics. We assume a working knowledge of category theory — categories, functors, adjunctions, monads, and enough fibrational language to read a subobject fibration without alarm — and we state the constructions at the level of generality at which they are true.

The second is the reader who wants to ship something. Every construction in this document is accompanied by a specification that a machine accepts. Where we say that a calculus is a GSLT, we give the **language!** block that presents it. Where we say a property is checkable, we give the syntax in which one writes the check. The formal sections and the programmable sections are not two treatments of one subject; they are one treatment, and the reader who skips either will find the other incomplete.

1.3 Roadmap

Part I develops the objects. Section 2 places the definition in the lineage of structured operational semantics and climbs a three-rung ladder from algebraic data types through universal algebra to domain-specific languages. Section 3 adds the interaction constructor and the interaction cut, and discriminates the resulting notion with three machines that fail to be instances of it. Section 4 adds the section and the wrappability condition, and tabulates how each of our running examples does or does not qualify.

Part II assembles the category. Section 5 shows why the obvious definition of a morphism does not survive contact with the canonical encodings, and rebuilds it with contexts as the primary structure.

Part III exhibits three constructions on that category: the cost monad in Section 6, the history monad in Section 7, and the generation of type systems in Section 8.

Part IV is the implementation. Section 9 presents Rholang 1.4 as a data definition language and a control flow language, and reduces the running examples to specifications a machine accepts.

Part I — Objects

2 Graph-structured lambda theories

2.1 The lineage: structured operational semantics

The presentation style we depend on is Plotkin’s [14, 15]. Before structured operational semantics, giving the meaning of a programming construct meant either a denotational assignment into a domain or an informal appeal to an abstract machine. Plotkin’s proposal was that one could give the behaviour of a language *syntactically*, by inference rules over the term structure, with the rules for a compound term given in terms of the rules for its parts. The presentation was finite, it was checkable, and — this is the part that matters here — it was *structural*: the shape of the rules followed the shape of the grammar.

Milner’s *Functions as Processes* [13] adopts this style and sets the standard by which such presentations should be judged. The paper gives the π -calculus by a grammar, a structural congruence, and a small family of reduction rules, and then encodes the λ -calculus into it. What makes the presentation exemplary is not merely that it is correct but that it is *complete in the relevant sense*: the grammar, the congruence, and the rules together are the whole of the definition. Nothing is left to the reader’s operational imagination. One can mechanically check a claimed reduction, and one can mechanically check that an encoding respects reduction, because there is nothing to the language beyond what has been written down.

Cardelli and Gordon’s presentation of the mobile ambient calculus [5] adopts precisely this style, and it is instructive to see how little has to change. The ambient calculus is a very different theory — its reductions restructure a spatial hierarchy rather than substitute a value — and yet the presentation is recognisably the same document: sorts, term formers, a structural congruence, a handful of reduction rules, and congruence rules closing reduction under the term formers. The style is not an artifact of process calculi. It is a general format for presenting a language.

The observation on which this document rests is that once a presentation is *that* regular, it is data. A grammar is a signature. A structural congruence is a set of equations. Reduction rules and congruence rules are a set of rewrites. These are not analogies; they are the same information under different typography. A GSLT is what you get when you stop treating an SOS presentation as a document to be read by a human implementer and start treating it as a structure to be consumed by a machine.

Remark 2.1 (Why “graph-structured”). The term-formation structure of a presentation makes terms into trees; the equations identify some of those trees; the rewrites connect the resulting classes by directed edges. The object of interest is therefore not a set of terms but a graph, and the theory is graph-structured in the sense that the algebraic structure and the reduction structure live on the same carrier. The framing of operational semantics in this algebraic, graph-enriched style follows Stay and Meredith [17] and Baez and Williams [2].

2.2 The definition

Definition 2.2 (Graph-structured lambda theory). A **graph-structured lambda theory** is a multi-sorted term theory $G = (\Sigma, E, R)$ consisting of

- a signature Σ of sorts and operators,
- a set E of equations, imposing a structural congruence $\equiv$, and

- a set R of rewrite rules.

The definition is deliberately thin. A GSLT is any multi-sorted rewrite theory; nothing in Definition 2.2 says that the theory is about computation, or that it has processes, or that anything interacts with anything. The enrichments of Sections 3 and 4 add exactly the structure that later constructions require, and the discipline of the present document is to add it only when a construction needs it.

2.3 Presentation and classifying theory

Two descriptions of the same object appear in the literature, and it is worth reconciling them once so that we may pass between them silently thereafter.

The *presentation* is the syntactic triple of Definition 2.2. It is what a developer writes and what a machine accepts.

The *classifying theory* of a presentation is a category $\mathbf{T}$ with finite limits and a chosen cartesian closed structure, equipped with its subobject fibration $\pi : \text{Sub}(\mathbf{T}) \rightarrow \mathbf{T}$, a distinguished object Pr of programs, and a distinguished subobject

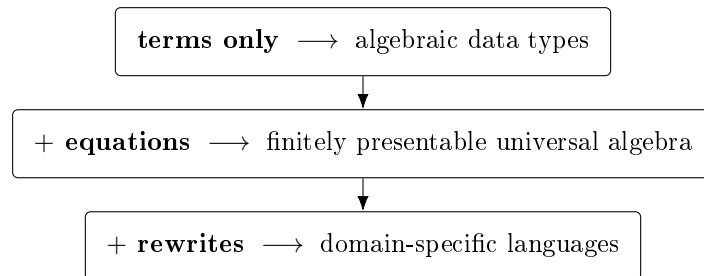
$$(\rightsquigarrow) \mapsto \text{Pr} \times \text{Pr}$$

of one-step reduction, together with entailments generating the base rewrites and their closure under the term formers. This is the presentation-independent form in which the type-theoretic construction of Section 8 takes its input.

We take the presentation as primary throughout, for the practical reason that it is what one can type; and we use the classifying theory when a construction is more naturally stated in the internal language. The passage from one to the other is the usual one and we do not belabour it.

2.4 The ladder

Definition 2.2 has three components, and the reader can switch two of them off. Doing so produces a ladder of three rungs, each of which is a familiar and independently useful thing. We climb it because it is the most efficient route we know from “I already know what this is” to “I did not know you could do that.”



2.4.1 Rung one: terms only

Take $E = \emptyset$ and $R = \emptyset$. What remains is a multi-sorted signature: a set of sorts and a set of operators with declared argument sorts and result sort. The terms over such a signature, up to nothing at all, are exactly the inhabitants of an algebraic data type.

```

language! {
  name: Json,

  types {
    Value
    Field
  },

  terms {
    JNull . Value ::= "null" ;
    JBool . b:Bool |- b : Value ;
    JNum . n:BigRat |- n : Value ;
    JStr . s:Str |- s : Value ;
    JArr . Value ::= "[" List(Value) "]" ;
    JObject . Value ::= "{" List(Field) "}" ;
    Field . k:Str, v:Value |- k ":" v : Field ;
  },

  equations { },
  rewrites { },
}

```

There is nothing here a working developer has not written a hundred times, in whatever notation their language provides for sums of products. The point of the rung is not that it is novel but that it is *included*: the specification format that will shortly describe the π -calculus describes a JSON document with no change of register.

2.4.2 Rung two: equations

Now populate E . The terms are quotiented by a congruence, and what one is presenting is no longer a free term algebra but an algebra satisfying equations — that is, a finitely presentable object of universal algebra. Monoids, groups, rings, lattices, semilattices, and the various bag- and set-like structures all live at this rung.

```

language! {
  name: Monoid,

  types { M },

  terms {
    Unit . M ::= "e" ;
    Mul . x:M, y:M |- x "*" y : M ;
  },

  equations {
    Assoc . |- (Mul (Mul X Y) Z) = (Mul X (Mul Y Z)) ;
    UnitL . |- (Mul Unit X) = X ;
    UnitR . |- (Mul X Unit) = X ;
  },

  rewrites { },
}

```

The rung matters to us for a specific downstream reason. The equational theory of a single operator — whether it is free, associative, or associative-commutative — turns out to determine whether proximity to a resource confers authority over it (Section 6.5), and whether an interaction

surface can be carried by position or must be named explicitly (Section 3.3). Equations are not decoration. They are the difference between an object-capability discipline and an ambient-authority leak.

2.4.3 Rung three: rewrites

Finally populate R . Now the data can wiggle.

This is the rung at which a specification stops describing a static structure and starts describing a language with a behaviour. The reduction rules of an SOS presentation go here, and so do the congruence rules that close reduction under the term formers, in exactly the shape Plotkin, Milner, and Cardelli–Gordon wrote them.

```
language! {
  name: Lambda,

  types { Term },

  terms {
    Lam . ^x.body:[Term -> Term] |- "lam " x "." body : Term;
    App . fun:Term, arg:Term      |- "(" fun "," arg ")" : Term;
  },

  equations { },

  rewrites {
    Beta . . |- (App (Lam fun) arg) ~> (eval fun arg);
    AppCongL . | M0 ~> M1 |- (App M0 N) ~> (App M1 N);
    AppCongR . | N0 ~> N1 |- (App M N0) ~> (App M N1);
    LamCong . | S ~> T   |- (Lam ^x.S) ~> (Lam ^x.T);
  },
}
```

Read the **rewrites** block against a textbook presentation of the λ -calculus and the correspondence is immediate: one base rule and three congruence rules, with the binder handled by the $\hat{x}$. notation and substitution by **eval**. The horizontal bar of an inference rule has become $|-$, and the premises sit to its left. This is not a translation of the λ -calculus into some other formalism. It is the λ -calculus, written so that a machine can read it.

Remark 2.3 (What the ladder buys). A reader arriving at rung three from rung one has crossed from data to computation without changing tools. That is the whole design argument for the format, and it is worth stating plainly: the reason to present a language as (Σ, E, R) is not elegance, it is that the same machinery that gives you a parser and a pretty-printer for your JSON documents gives you a reduction engine for your calculus, and everything in Part III on top of that.

2.5 A worked presentation: the rho calculus

We will need the rho calculus [9] as a running example, since it is both the host language of Part IV and one of the more instructive instances of the constructions of Part III. It is a reflective higher-order calculus: names are not generated by a restriction operator but are *quoted processes*, and the quoting and unquoting operators $@$ and $*$ mediate between the two sorts.

```
language! {
  name: RhoCalc,

  types { Proc Name },
```



```

terms {
  PZero . | - "Nil" : Proc;
  PDrop . n:Name | - "*" n : Proc;
  POutput . n:Name, q:Proc | - n "!" "(" q ")" : Proc;
  PInput . n:Name, ^x.p:[Name -> Proc] | - "for" "(" x "<-" n ")" p : Proc;
  PPar . ps:HashBag(Proc) | - "{" ps.*sep("|") "}" : Proc;
  NQuote . p:Proc | - "@" p : Name;
},

equations {
  QuoteDrop . | - (NQuote (PDrop N)) = N ;
},

rewrites {
  Comm . | - (PPar {(PInput n ^x.p), (POutput n q), ...rest})
    ~> (PPar {(subst ^x.p (NQuote q)), ...rest});
  Drop . | - (PDrop (NQuote P)) ~> P;
  ParCong . | S ~> T | - (PPar {S, ...rest}) ~> (PPar {T, ...rest});
},
}

```

Three features of this presentation will be load-bearing later. The parallel composition **PPar** takes a **HashBag**, so its equational theory is associative-commutative — position among parallel components carries no information. The **Comm** rule matches two components of that bag and leaves the remainder **...rest** untouched, which is what makes it a rule about a *site* of interaction rather than about whole terms. And the quote constructor **NQuote** gives the theory a way to name its own programs, which will matter when we ask whether a theory can internalise apparatus applied to it.

3 Interactive GSLTs

3.1 Naming the site of interaction

A GSLT is any multi-sorted rewrite theory. Most of what we want to do requires more: we want to know *where* the theory's programs meet whatever they compute against.

Definition 3.1 (Interactive GSLT). A GSLT $G = (\Sigma, E, R)$ is **interactive** when it carries, over and above the triple:

- a distinguished *interacting sort* P of programs;
- a binary *interaction constructor* $K(P, E)$, representing the interaction of a program with an environment E — an operand of the interacting sort, with $E = P$ in the process-calculus instances;
- at least one base rewrite rule in R whose left-hand side features K — the *interaction rule* — generating a labelled transition system on P modulo $\equiv$ on which bisimulation is the intended equivalence.

Definition 3.2 (The category **iGSLT**). **iGSLT** is the category whose objects are interactive GSLTs and whose morphisms are theory maps — sort- and operator-preserving translations respecting $\equiv$ and R — that are bisimulation-preserving on the interacting sort.

The constructor K and the K -headed rule are exactly what a bare GSLT lacks. A GSLT is any multi-sorted rewrite theory; an interactive GSLT *names a site of interaction and a rule that*

fires there. The motivating instances span calculi with and without binding: CCS [12] with P the processes, $K = |$ and interaction by complementary actions; the rho calculus with $K = |$ and $R \ni \text{COMM}$; the λ -calculus with P the terms, $K = \text{App}, \equiv = \alpha$ and $R \ni \beta$; and the interaction categories of Abramsky, Gay and Nagarajan [1], where K is composition along a shared interface.

3.2 The interaction cut

Definition 3.1 asks only that *some* rule mention K . In every instance we care about, that rule has a great deal more shape, and naming the shape is what makes the constructions of Part III possible. Write the generating rule of R as

$$\frac{(C'_p, C'_e) = \text{compute}(I, J, C_p, C_e)}{K(K_p(I, C_p), K_e(J, C_e)) \longrightarrow K'(C'_p, C'_e)} \quad (\dagger)$$

distinguishing:

the interaction constructor K ,

which brings two operands into contact. In rho and CCS, $K = |$; in λ , $K = \text{App}$.

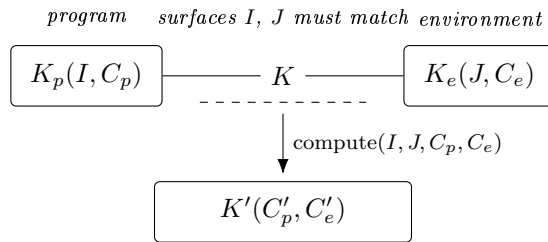
two introductions $K_p(I, C_p)$ and $K_e(J, C_e)$,

each a constructor separating an *interaction surface* (I, J) — the part that must match for the cut to fire — from a *continuation*: C_p the program continuation, the rest of the computation, and C_e the environment continuation, the rest of the environment.

the contraction compute ,

a partial operation that, on a surface match, combines the two continuations into residuals C'_p, C'_e repackaged by K' .

The factoring is Milner's [13]. In the polyadic π -calculus the input for $(y \leftarrow x)P$ is the pair $(x, \lambda y.P)$ and the output $x!(u).Q$ is the pair $(x, [u, Q])$: the subjects x are the surfaces that must match, the objects $\lambda y.P$ and $[u, Q]$ are the continuations, and the contraction is Milner's *pseudo-application*, returning $P\{u/y\} \mid Q$.



We call $(\dagger)$ an *interaction cut*: two introductions meet along matching surfaces and a contraction combines their continuations. Both sides are structured — the eliminand is not a bare term but an introduction carrying its own continuation.

Remark 3.3 (Binding is not the point). Pseudo-application factors into two parts: the substitution $P\{u/y\}$, by which a datum migrates from the environment continuation into the program continuation through a binder, and the release $| Q$ of the environment's residual. Binding is therefore *one mode by which information migrates* under the contraction, not an intrinsic feature of the cut. CCS realises the same shape with no migration at all, and is in this sense the paradigmatic instance.

3.3 Surfaces: nominal and structural

A surface may be carried in either of two ways, and which one a calculus uses is dictated entirely by the equational theory of K .

It may be carried *nominally*, as an explicit I or J that must match — as the subject x does in ρ and π , or as complementary labels do in CCS. Or it may be carried *structurally*, by the rigidity of K : when K is free, carrying no equations at all, position itself is unforgeable and the context $K([], -)$ is the surface. A nominally degenerate — absent — surface is then not a missing match condition but one carried by the geometry of K instead.

The λ -calculus is exactly this case on the environment side. App is free, and its rigidity supplies the surface that no explicit J records. Each application comprises a location together with a function and an argument; the location is the surface, the function is the program and the argument is the environment — or dually. A variable, we note, is not a surface, though it is closely related to one.

The dividing line is *freeness*, not commutativity. An equation on K is a congruence step: it rewrites adjacency without firing a cut. Associativity re-brackets adjacency, a unit law inserts and deletes neighbours, idempotence copies them, and associative–commutativity dissolves position altogether into a freely mixed soup. Each forges position, and only a free K admits no such move. We return to the consequences in Section 6.5.

3.4 Discriminating examples: three machines that are not interactive

It is easy to accumulate examples of a definition and hard to see what the definition excludes. We therefore work three non-examples in detail. They are not exotic: they are the machines with which most readers first learned what computation is.

3.4.1 Turing machines

A Turing machine is unproblematically a GSLT. Its sorts are states, tape symbols, and configurations; its operators build a configuration from a state, a tape, and a head position; its rewrites are the transition table together with the mechanics of moving the head. One can write the presentation down and a machine will accept it. We do so in Section 9.3.2.

What it is not is *interactive* in the sense of Definition 3.1. To make it interactive one must exhibit an interaction constructor $K(P, E)$ and a K -headed rule. The only plausible candidate for the site of interaction is the contact between the finite control and the tape: that is, after all, where the machine’s steps happen. But the definition asks that E be an operand of the interacting sort, and the tape is not of the sort of the control. The candidate cut is heterogeneous, and $K(P, E)$ has no well-sorted instance.

Non-example 3.4 (Turing machines). The Turing machine has no naive presentation as an interactive GSLT. The cut would be between automaton and tape, which are of different kinds.

The failure is worth dwelling on because it is not a failure of expressive power. A Turing machine can compute anything a ρ calculus term can compute. What it lacks is a *site*: a place in the syntax where two things of the same kind are brought into contact, such that one could ask what else might have been brought into contact there instead. The absence of such a site is precisely the absence of an environment in which the machine could be placed — which is another way of saying that a Turing machine is a model of a closed computation, and interactivity is about open ones.

3.4.2 Moore and Mealy machines

The transducers fail in the same way and differ from each other in an instructive second respect.

A Moore machine emits an output determined by its current state alone; a Mealy machine emits an output determined by its current state *and* the input symbol just consumed. Both are automata reading a stream, and for both the only candidate cut is between the machine and the stream — again heterogeneous, again no well-sorted $K(P, E)$. On the first axis, then, the three machines fail identically, and one might think a single non-example would have sufficed.

It does not, because the pair discriminates along a *second* axis: the migration classification of Remark 4.7 below. Were one to force an interaction-cut reading on these machines, the Moore machine's output is a function of the state alone, so nothing of the environment reaches the emitted value; the contraction would be pure release, the null-migration corner occupied by CCS. The Mealy machine's output depends on the consumed symbol, so the environment's datum does reach the residual; the contraction would migrate. Two machines that fail interactivity for the same reason nevertheless sit at different points of the classification that organises the successes.

Non-example 3.5 (Moore and Mealy machines). Neither transducer has a naive presentation as an interactive GSLT, for the reason of Non-example 3.4. Under a forced reading they would occupy the null-migration and binding-migration positions respectively.

3.4.3 Presentation versus encoding

The reader will object that Turing machines can obviously be *run* inside a process calculus, and that the objection above is therefore pedantic. The objection is correct and the distinction it presses on is the important one.

There is a difference between a theory *having* a presentation and a theory *admitting an encoding into* one that does. Encoding a Turing machine into the rho calculus — reifying the tape as a collection of processes communicating on channels, and the head as a process that moves among them — produces a rho term whose behaviour simulates the machine. That is a genuine and useful construction, and we exhibit it in Section 9.3.2. But the result is a statement about rho, not about Turing machines. The interaction sites in the encoded term are rho's sites, and the questions one can ask of them are questions about rho contexts.

This distinction is the entire subject of Section 5, where the morphisms of the category **GSLT** turn out to be exactly the well-behaved encodings, and where the conditions under which an encoding tells you something about the source rather than merely about the target are isolated.

4 Continued interactive GSLTs

Interactivity names a site. The constructions of Part III need more: they need to be able to *intervene* at the site — to interpose a token, to record an event, to attach a guard. Three further conditions suffice, and this section isolates them.

4.1 Section-equipped

The first condition concerns the relationship between a term and its congruence class. Let T be the free syntax of the interacting sort and $\hat{T} = T/\equiv$ its quotient by structural congruence. Two maps relate them:

- the *quotient* $T \rightarrow \hat{T}$, respecting $\equiv$, invertible up to congruence. This is the structure a name constructor exposes when channels are taken up to $\equiv$ — in rho, $@P$ with $@P = @Q \iff P \equiv Q$.

- a *section* $\text{cf} : \widehat{T} \rightarrow T$, a choice of canonical representative per class.

Definition 4.1 (Section-equipped). An interactive GSLT is **section-equipped** when it comes with a computable section $\text{cf} : \widehat{T} \rightarrow T$ of its $\equiv$ -quotient — equivalently, a computable canonical-form function on the interacting sort.

The condition is sharper than it looks, and the λ -calculus is the test that shows why. The λ -calculus has no analogue of the quote constructor $@$. It passes anyway, because what is required is not a channel constructor but a section: for λ the congruence is α and a canonical representative of each α -class is the de Bruijn encoding [6]. Rho fuses the quotient and the section into one operator viewed two ways; λ pulls them apart; the requirement is identical in both.

4.2 Wrappability

The second condition asks that the contraction tolerate having its operands decorated. Anticipating Section 6, write $\mathbf{T}$ for a sort of *wrapped* terms.

Definition 4.2 (Wrappability). The contraction compute is **wrappable** when it is well-sorted as an operation on wrapped continuations, $\text{compute} : \mathbf{Surf} \times \mathbf{Surf} \times \mathbf{T} \times \mathbf{T} \rightarrow \mathbf{T} \times \mathbf{T}$ — equivalently, when compute is definable on decorated continuations, mapping decorated inputs to decorated outputs.

The condition is mild for substitution-like contractions: substituting a wrapped term into a position yields a wrapped term, so compute preserves the sort for free and need not know that wrapping exists. It excludes contractions that inspect and dismantle their operands — an opaque contraction returning bare residuals would violate it.

4.3 The definition

Definition 4.3 (Continued interactive GSLT). A **continued interactive GSLT** is an interactive GSLT equipped with:

- (i) a presentation of its dynamics in interaction-cut form ($\dagger$), naming K, K_p, K_e, K' and the partial contraction compute ;
- (ii) a section (Definition 4.1); and
- (iii) wrappability (Definition 4.2).

We write **ciGSLT** for the resulting category, whose morphisms are theory maps that are bisimulation-preserving on the interacting sort and *quote-faithful*: injective on the reachable signature keys.

The adjective names a strict enrichment. Clauses (i)–(iii) are data and conditions added on top of an interactive GSLT, not a restriction of it; forgetting them recovers the underlying interactive GSLT.

Definition 4.4 (The forgetful functor). $U : \mathbf{ciGSLT} \rightarrow \mathbf{iGSLT}$ sends a continued interactive GSLT to its underlying interactive GSLT, discarding the cut presentation, the section, and the wrappability witness, and acts as the identity on the underlying theory map of a morphism.

Proposition 4.5 (The distinction is proper). *U is faithful but neither full nor essentially surjective. Hence $\mathbf{ciGSLT}$ is a genuine, proper subcategory of $\mathbf{iGSLT}$: strictly more structure on objects, strictly fewer morphisms between them.*

Instance	K	E on K	I, J	C_p / C_e	compute (migration)
Calculator	—	—	—	—	no interaction rule
CCS		AC	$a, \bar{a}$	P / Q	pure release $P \mid Q$ (none)
Interaction cats.	o	—	shared interface	proc. / proc.	interface composition (none)
rho, π (sync)		AC	x, x	$\lambda y.P / [Q_1, Q_2]$	pseudo-app. $P\{Q_1/y\} \mid Q_2$ (binding)
rho, π (async)		AC	x, x	$\lambda y.P / [Q, -]$	pseudo-app. $P\{Q/y\}$ (binding)
λ	App	free	$x, -$	$\lambda x.M / N$	application $M\{N/x\}$ (binding)
Ambients		AC	n, n	P / Q	boundary dissolve/-move (spatial)
Turing	<i>none</i>	—	—	—	control/tape heterogeneous
Moore	<i>none</i>	—	—	—	control/stream heterogeneous
Mealy	<i>none</i>	—	—	—	control/stream heterogeneous

Table 1: Interaction-cut data. A surface entry of — denotes a nominally degenerate surface carried structurally by the rigidity of K ; an environment-continuation entry of — denotes a degenerate continuation, which is what distinguishes asynchronous from synchronous output.

Discussion. Faithful, not full. U forgets no information about a morphism’s action, so it is faithful. It is not full because a **ciGSLT**-morphism must additionally be quote-faithful: an **iGSLT**-morphism that is bisimulation-preserving yet collapses two distinct signature keys is a map between the underlying objects with no lift.

Not essentially surjective. Clause (i) requires the dynamics to factor as contact then contraction, which a theory with non-local or unfactorable rewrites need not; clause (ii) requires a computable section, which a theory with undecidable structural congruence lacks; clause (iii) requires the contraction to preserve wrapping. An interactive GSLT failing any of these has no preimage. $\square$

Remark 4.6 (λ exhibits the distinction). The λ -calculus is an object of **iGSLT** as such: terms, α, β . It becomes an object of **ciGSLT** only once one *chooses* the added structure — the de Bruijn section for (ii), the degenerate- K_e presentation of β for (i), substitution-on-thunks for (iii). The same underlying calculus sits at both levels, witnessing that “continued” is added structure and not a property read off the interactive GSLT alone.

4.4 How the examples are examples

Remark 4.7 (The migration spectrum). The instances are usefully ordered by how much information the contraction moves from the environment continuation into the program continuation. Four modes occur: no migration (CCS, interaction categories), migration by binding (rho, π , λ), migration by spatial restructuring (ambients), and migration as interface composition.

Table 1 records the interaction-cut data of each running example, and Table 2 records whether that data satisfies the successive conditions. The final column of Table 2 is the discrimination the two enrichments were introduced to make.

Instance	GSLT	has K -rule	section cf	wrappable	Verdict
Calculator	✓	—	normal form	—	GSLT only
CCS	✓	✓	normal form for AC	✓	continued interactive
Interaction cats.	✓	✓	interface normal form	✓	continued interactive
ρ, π (sync)	✓	✓	normal form for AC	✓	continued interactive
ρ, π (async)	✓	✓	normal form for AC	✓	continued interactive
λ	✓	✓	de Bruijn	✓	continued interactive*
Ambients	✓	✓	normal form for AC	✓	continued interactive
Turing	✓	—	—	—	GSLT only
Moore	✓	—	—	—	GSLT only
Mealy	✓	—	—	—	GSLT only

Table 2: Verdicts. * λ qualifies only once the added structure is chosen; see Remark above. The three machines are perfectly good GSLTs and fail at the first enrichment, not the second.

Two readings of the tables are worth making explicit.

First, the calculator row and the three machine rows fail for *different* reasons, and the tables record the difference. The calculator has no interaction rule because it has no notion of a program meeting an environment at all — it is a language of closed expressions. The machines have a notion of the control meeting something, but the something is of the wrong sort. Both end at “GSLT only,” and a reader who wanted to know which of the two situations obtains would have to look at the K column.

Second, λ is the only row whose verdict carries a caveat, and the caveat is the content of Proposition 4.5. Nothing about the λ -calculus as ordinarily presented determines a section; de Bruijn is a choice, and a different choice — a different canonical form for α -classes — would give a different object of **ciGSLT** over the same object of **iGSLT**. This is what it means for U to fail to be essentially surjective on the nose while being surjective on the underlying calculi one cares about.

Part II — The category

5 The category GSLT

Part III consists of constructions that take a theory and produce a theory. To say anything about such constructions — that they are functorial, that they are monads, that they have adjoints — we need a category of theories. This section builds it, and the build is less routine than one might expect: the obvious definition is wrong, and the way it is wrong is instructive.

5.1 The naive definition and three defects

The obvious move is to take objects to be GSLTs and morphisms to be signature-preserving translations that preserve bisimulation. Three things go wrong.

The constant map. Send every term to 0. This preserves signatures trivially and preserves bisimulation trivially, since the image has no transitions to fail to match. It is a morphism from anything to anything, and its presence means the category orders nothing.

Milner’s encoding is not a map of terms. The canonical encoding of the λ -calculus into the π -calculus [13] — the very construction that established the modern practice of comparing calculi — sends a λ -term not to a π -term but to a π -term *parameterised by a name*: $\llbracket M \rrbracket_u$, the process that realises M and reports its result on u . Signature preservation does not typecheck, because **App** is not sent to any single π operator; it is sent to a construction involving restriction, parallel composition, and output. Requiring morphisms to be signature homomorphisms excludes the motivating example.

Ambient contexts observe too much. Suppose one encodes π into rho. A rho context may quote a process, compare quotations, and generally observe the syntactic form of what it is handed — capabilities no π context has. Bisimulation computed over *all* rho contexts will therefore separate the images of π -terms that no π context can separate. Under the naive definition, no interesting encoding is ever bisimulation-preserving, for a reason that has nothing to do with the encoding.

5.2 Contexts as the primary structure

The three defects have one cure. The first says we need a non-degeneracy condition; the second says terms are the wrong primitive; the third says the observer must be restricted to what the source can express. Taking contexts rather than terms as primary addresses all three at once, because a term is a nullary context, and because the observers of a term are exactly the contexts one can place around it.

A theory presents a symmetric multicategory of contexts. Its objects are *interfaces*, recording the sort of a hole, its binding stage, and its interaction surface. Its multimorphisms are contexts with holes of the given interfaces and a result of the given interface; composition is plugging. Terms are the nullary contexts.

Transitions are then labelled by contexts: rather than $P \rightarrow P'$, we record $P \xrightarrow{C} P'$, meaning that placing P in the context C enables the step. Bisimulation is taken over these context-labelled transitions.

Definition 5.1 (Morphism of theories). A morphism of GSLTs is a pseudofunctor of context multicategories that is bisimulation-preserving for context-labelled transitions, where the bisimulation on the target is computed only over the image of the source’s contexts.

Each clause repairs one defect. Because the map is defined on contexts and terms are nullary contexts, an encoding like Milner’s — which sends a term to a context-parameterised process — is expressible: the parameterisation is the interface. Because bisimulation on the target is computed over the image, the ambient-observation defect disappears by construction; the encoded π -terms are compared by the encoded π contexts, which is the comparison one meant to make. And the constant map is excluded by the non-degeneracy conditions of the next subsection.

Definition 5.2 (The category **GSLT**). **GSLT** is the category whose 0-cells are GSLTs and whose morphisms are the maps of Definition 5.1: bisimulation-preserving for the context-labelled transition relation, which is thereby a congruence.

This is the official definition for the remainder of the document. The additional conditions imposed by particular constructions — quote-faithfulness for the cost functor of Section 6, preservation of Pr and $\rightsquigarrow$ for the type-system functor of Section 8 — are stated where those constructions need them, and are refinements of Definition 5.1 rather than rivals to it.

5.3 Hosting and exhausting

Two conditions prevent the comparison from being trivial.

Definition 5.3 (Hosting). A morphism $F : G \rightarrow H$ is *hosting* (faithful) when the target can run the source: the encoding reflects as well as preserves the context-labelled transition structure, so that no branching present in G is lost in H .

Definition 5.4 (Exhausting). $F : G \rightarrow H$ is *exhausting* (dense) when the target is nothing the source cannot assemble: every context of H relevant to the image is, up to the equivalence, in the image of a context of G .

The constant map fails hosting immediately. Together the conditions induce a preorder on theories whose degrees *refine* Turing completeness rather than replacing it: the classical notion is recovered as the trace collapse, obtained by coarsening context-labelled bisimulation all the way to the input/output relation.

Remark 5.5 (Why not Turing completeness). Expressiveness comparisons between computational models are conventionally anchored to Turing completeness. For functions this works. For interactive systems it does not, because Turing completeness is a statement about the input/output relation, and the input/output relation is the coarsest invariant in the linear-time/branching-time spectrum [18], while the equivalences that matter for interaction sit at the other end. Calculi of identical Turing power are routinely separated by uniform-encoding criteria, so the yardstick is measuring something other than what is being compared.

The obvious repair — demand behavioural rather than computational universality — merely relocates the problem, since the arbiter of which behavioural equivalence to use is an adequate modal logic, and the logical apparatus available in the ambient setting is inherited from the λ -calculus. An absolute notion is quietly λ -relative. Definition 5.1 replaces the yardstick with a relative property: fix a probe and ask what that probe can see.

Remark 5.6 (A note on complexity). The same preference for intensional over extensional comparison has a consequence for complexity theory that we record without pursuing. Equivalence of regular languages is PSPACE-complete, while the corresponding bisimulation problem is polynomial; equivalence of context-free languages is undecidable, while the corresponding bisimulation problem is tractable. We want the witnesses, not the extensions they give rise to. Infinitary objects like languages cannot be grasped directly, so an intensional representation is needed — and once one has an intensional representation, one has a bisimulation problem rather than a language-comparison problem.

Part III — Constructions

The three sections that follow have the same shape. Each takes a theory presented as an interaction cut and produces another such theory, decorated with apparatus. Each is a monad. Each resolves through a free/forgetful adjunction that installs and removes the apparatus. And in each case, when the base theory is sufficiently expressive, the apparatus can alternatively be encoded back into the base — so that the base performs the metering, or the logging, or the checking, with its own computation. We note that possibility where it arises and do not develop it.

6 The cost monad

6.1 What is being built

Given a continued interactive GSLT G , the cost endofunctor $\mathcal{C}$ produces a calculus $\mathcal{C}(G)$ in which every interaction is gated on the consumption of a token. The design question is where the gating lives, and the answer that makes the construction work is: *in the grammar of the image*.

In $\mathcal{C}(G)$ the functor adjoins a sort **Sig** of signatures, a wrapped-term sort **T** with the wrapper $\{\cdot\}_s : P \times \mathbf{Sig} \rightarrow \mathbf{T}$, and a sort **Stk** of token stacks

$$S ::= () \mid s :: S, \quad s \in \mathbf{Sig}.$$

The pivotal discipline is grammatical: in $\mathcal{C}(G)$ every continuation slot of K_p and K_e has sort **T**. A continuation in the metered calculus is therefore not a bare process awaiting metering; it is a *metered thunk* $\{P\}_s$, whose activation is gated by a token keyed to s , and whose own continuations are again thunks.

The signature constructor is $\# = \text{digest} \circ \text{cf}$: the section of Definition 4.1 chooses a canonical representative, and a collision-resistant digest commits to it. Note that $\#$ does *not* respect $\equiv$, and must not, since a commitment identifying congruent-but-distinct representatives would be no commitment. Signatures never reduce; they are a rewrite-free sub-theory meeting the behavioural theory only at the matching guard, so the $\equiv$ -incoherence of $\#$ is harmless for bisimulation, which factors as “ P -bisimulation gated by key matching.”

6.2 The gated rules

A wrapped redex is forced by consuming a matching token. The single-signature rule seals the whole redex under one s and funds it from a purse located at the surface $L = \text{near}(I, J)$:

$$\frac{L = \text{near}(I, J) \quad (\{C'_p\}_{t_1}, \{C'_e\}_{t_2}) = \text{compute}(I, J, \{C_p\}_{t_1}, \{C_e\}_{t_2})}{K(\{K(K_p(I, \{C_p\}_{t_1}), K_e(J, \{C_e\}_{t_2}))\}_s, S(L, s :: p)) \longrightarrow K(K'(\{C'_p\}_{t_1}, \{C'_e\}_{t_2}), S(L, p))} \quad (\text{R1})$$

The remaining cases seal the two interaction surfaces rather than the whole redex, funding per surface, and are what the associative-commutative structure of $|$ requires when a redex is split across the bag or its tokens are supplied separately. We write them as (R2) and (R3) and do not reproduce them here.

The decisive point is what is *absent*. There is no re-wrapping step and no lifted contraction. By the typing of compute , the residual $K'(\{C'_p\}_{t_1}, \{C'_e\}_{t_2})$ is already built from wrapped terms.

Definition 6.1 (Well-wrapped). A configuration of $\mathcal{C}(G)$ is *well-wrapped* when it is well-sorted in the grammar of $\mathcal{C}(G)$ — in particular, every continuation occupies a slot of sort **T**, so every redex occurrence lies inside a wrapper.

Lemma 6.2 (Subject reduction for wrapping). *Well-wrapping is preserved by (R1)–(R3).*

Proof sketch. Each rule replaces a wrapped redex by $K'(\{C'_p\}_{t_1}, \{C'_e\}_{t_2})$ with $(C'_p, C'_e) = \text{compute}(\dots)$. Since compute lands in $\mathbf{T} \times \mathbf{T}$ and K' packages terms of sort **T**, the residual is well-sorted; the consumed cell leaves a well-sorted stack; redexes elsewhere are untouched. $\square$

Corollary 6.3 (No leak, by construction). *In a well-wrapped configuration no redex can fire without being unwrapped, i.e. without consuming a token. There is no dynamic wrapping operation; the cost-accounting steps only ever unwrap.*

Remark 6.4 (Eager versus lazy metering). An alternative discipline re-wraps the contractum at contraction time, traversing it to sign each exposed redex — charging eagerly for every redex a step exposes. Wrapping by construction charges lazily: each redex is born wrapped and is charged only when actually forced. Lazy metering pays for work performed, not work merely exposed; redexes beneath a discarded binder are never charged. The token stack drains exactly in step with the reductions performed, which is also the operational reality of fuel consumption.

Remark 6.5 (Duplication needs no fresh signatures). Because multiplicity is carried by the stack and not by key distinctness, copying a wrapped term $\{Q\}_s$ copies its key but not its tokens: n copies of an s -redex require n cells keyed to s . Created redexes — where a substituted wrapped term lands in head position — are guarded for the same reason, since the substituted material carries its own wrapper. One mechanism handles duplicated and created redexes alike.

6.3 Two monoids: space and time

The construction places two combination operators side by side, of different character.

The interaction constructor K is *spatial*: it records what is in contact with what, and may carry equations — associative–commutative in rho, free in λ .

The cons operator $::$ is *temporal*: it records what is spent next, then next. It is a free monoid, never commutative, because reduction is sequential even when interaction is parallel. Each forcing pops a cell; the order of popping is the order of steps. The stack is the time axis of the computation reified as a term: $\equiv$ leaves it invariant, $\longrightarrow$ pops it, and its ordering is the arrow of the computation.

Proposition 6.6 (Stack consumption is the modulus). *For a terminating reduction in $\mathcal{C}(G)$, the number of cells consumed equals the number of forced redexes, which equals the number of cuts eliminated — including those introduced by duplication, each charged when forced. The consumed prefix of the temporal stack is an exact operational modulus for the cut elimination, realised by running the reduction rather than by a separate traversal.*

6.4 The monad and its resolution

Construction 6.7 (Unit and multiplication). Define $\eta_G(P) = \{P\}_{()}$, wrapping each term with the unit signature. Since $()$ is the unit of the signature monoid, a $()$ -keyed redex fires without net resource, so η_G embeds G as the cost-free fragment of $\mathcal{C}(G)$. Define $\mu_G : \mathcal{C}^2(G) \rightarrow \mathcal{C}(G)$ by multiplying nested signatures, $\{\{P\}_{s_2}\}_{s_1} \mapsto \{P\}_{s_1*s_2}$, and concatenating the stack-of-stacks into a single stack.

Proposition 6.8 (The cost monad). *$(\mathcal{C}, \eta, \mu)$ is a monad on **ciGSLT**. Its laws descend from the laws of the two constituent monoids: the unit laws from $s * () = s = () * s$ together with the singleton/empty laws of the list monad, and associativity from associativity of $*$ and of concatenation.*

Remark 6.9 (A genuine, non-idempotent resource monad). $\mathcal{C}$ is not idempotent: μ_G is no isomorphism, because concatenating two stacks forgets the boundary between them and multiplying two grades forgets the factorisation. Flattening is a true merge of two resource accounts, not the collapse of a redundant copy. This is the expected signature of a resource monad, and it distinguishes $\mathcal{C}$ from idempotent closure monads.

Proposition 6.10 (Free–forgetful resolution). *Let **Cost-ciGSLT** be the category of cost-accounted continued interactive GSLTs. The functor **Install** that adjoins the apparatus is left adjoint to the functor **Forget** that strips it, and the induced monad on **ciGSLT** is $\mathcal{C}$.*

The embedding is structural and strict: Install adjoins sorts and rules, Forget erases them on the nose. Crucially, forgetting is not behaviour-preserving in the metering sense — erasing the apparatus removes the gating, so the forgotten theory runs its interactions without friction. The resolution answers “what apparatus was added?” and lets us remove it; it does not claim the metered and un-metered dynamics agree.

Remark 6.11 (Internalisation). When the base theory is sufficiently expressive — enough to code its own higher-order syntax, whether by binding or by reflection — the entire metering apparatus is itself expressible in the base. Tokens become encoded data and the gated rules become an interpreter loop, so that the base performs the metering with its own computation. This is a second and quite different erasure from forgetting: rather than stripping the apparatus it dissolves it into the base. We record its availability and do not develop it here.

6.5 Capabilities and located purses

A cost-accounted calculus is a calculus in which the right to draw on a resource stack is itself an authority. We must therefore ask what confers that authority, and ensure the construction does not silently grant it. The answer turns entirely on the equational theory of K , in the manner already anticipated in Section 3.3.

Free K : position is the capability. When K carries no equations, nothing can drift into contact with a stack it does not already neighbour. A capability to draw on a stack S is then literally the context $K([], S)$ — the right to occupy the hole adjacent to S . Because K is rigid, occupying that position is the whole of the authority, and confinement is automatic. This is the λ -calculus regime and it is the object-capability ideal: authority is structural, held by reference, and unforgeable by construction.

Non-free K : the leak. The moment K carries any equation, position is no longer exclusive. An equation is a congruence step that changes adjacency without firing a cut, so it lets a program drift into contact with a stack it did not structurally neighbour. Associative–commutativity is the limit: the bag is a freely mixed soup, every program adjacent to every stack, and rho lives exactly here. Taking proximity as authority in that regime is precisely *ambient authority* [11], and it is a leak.

The repair: nominal surfaces and located stacks. To recover an object-capability discipline under a non-free K , carry the surface nominally and gate on a nearness operator $\text{near}(I, J)$, which when defined returns the surface at which the two meet. Interaction is licensed not by position but by the matchability of the surfaces. Then locate the stack at a surface,

$$S(I, s :: S'),$$

so that the cookie jar carries a label naming which hands may reach in. Authority over a resource is thereby named in the term rather than conferred by the geometry of the bag.

Remark 6.12 (Nearness bounds but does not eliminate competition). $\text{near}(I, J)$ does not by itself make access exclusive: two receivers may both be near the same surface. What it guarantees is that only holders of a matching surface may compete at all — it replaces “anyone adjacent” with “anyone holding a matching surface.” Exclusivity, where required, is a matter for the type discipline of Section 8.

7 The history monad

7.1 Keeping the past

A rewrite theory that discards its past is irreversible precisely because a configuration may have many predecessors. Recording which step was taken restores a unique one. That is the whole idea, and packaging it gives the mirror image of Section 6.

Definition 7.1 (The history endofunctor). $\mathcal{H}$ sends a continued interactive GSLT G to the theory whose configurations are pairs of a configuration of G and a word in the free monoid of rewrite events, with each rewrite additionally appending its own event to the word.

$\mathcal{H}$ is the writer monad over the free monoid of rewrite events. Its unit attaches the empty history; its multiplication concatenates.

Proposition 7.2. *$(\mathcal{H}, \eta, \mu)$ is a monad on $\mathbf{ciGSLT}$, and the evident functor installing the log is left adjoint to the functor stripping it, with induced monad $\mathcal{H}$.*

7.2 The free history is the reversible tree

Proposition 7.3 (Universal cover). *The free history makes the graph of computation paths out of a term into a tree: it is the universal cover of the reduction graph, and reversibility is exactly its unique-parent property.*

This is the structural content of the construction and it explains the erasures. An erasure of history is a monad quotient, and these form a lattice between the reversible cover and the irreversible base. The injective erasures compress the log without folding the tree; the identifying ones fold it, closing loops so that paths revisit states. Such an intermediate object is a quotient of the reduction graph's fundamental group by a chosen set of loops — an intermediate cover — with the order-forgetting (Mazurkiewicz) erasure folding exactly the commuting concurrent steps.

7.3 The ledger law

Underneath the covering-space picture sits an invariant needing no cohomology. Write σ for the fuel remaining and κ for the fuel recorded as spent. Then

$$\sigma + \kappa = \sigma_0,$$

initial fuel equals fuel remaining plus fuel recorded spent: a potential-plus-kinetic invariant of a cost-accounted, history-keeping run. Its one-way breakage under erasure is Landauer's principle [8] — the history one may erase for free is exactly the redundant, state-recoverable part.

7.4 The duality

The two constructions are ledgers of the same time axis, read in opposite directions.

	Cost $\mathcal{C}$	History $\mathcal{H}$
adjoined structure	token stack	event word
monoid	free (list), consumed	free (list), accumulated
behaviour along $\longrightarrow$	drains	accrues
records	what remains	what was spent
free object	fully funded run	universal cover
erasure lattice	grades of funding	subgroups of π_1
physical reading	bounded causality	recorded causality

A GSLT gives causality. $\mathcal{C}$ gives *bounded* causality. $\mathcal{H}$ gives the ability to *record* bounded causality. Running $\mathcal{C}$ on an $\mathcal{H}$ -equipped model gives the cost of making the recording, and the levels are worth keeping distinct rather than collapsing: the stacking $\mathcal{C}\mathcal{H}\mathcal{C}\mathcal{H}\dots$ is meaningful, and its fixed point is an object of interest we do not pursue here.

Remark 7.4 (Internalisation, again). As in Remark 6.11, a sufficiently expressive base theory can encode the logging apparatus into itself, performing the record-keeping with its own computation. The same caveat applies: this is a different erasure from forgetting, and we note rather than develop it.

8 Generating type systems: OSLF and the Hypercube

8.1 OSLF as a family

Operational Semantics in Logical Form (OSLF) is not a single algorithm but a *family* of them. Each member takes a theory presented as (Σ, E, R) together with a specification of a target logic, and generates from the two an assignment of predicates to terms. The lineage runs through Hennessy–Milner logic [7], Stirling’s modal characterisations, and, for the reflective setting, the namespace logic of the rho calculus [10].

The member we develop here is the *Hypercube* algorithm of Stay, Meredith and Wells, which generates not a modal logic for observing programs but a family of type systems for classifying them. Both directions matter, and it is worth saying why they are the same construction. A type is a predicate on terms; a modal formula is a predicate on terms; the difference is bookkeeping about what one intends to do with the predicate. What the family provides is the machinery for reading a rewrite rule as a formation rule for predicates.

8.2 Adequacy is not automatic

Because the difference between the two directions is bookkeeping, it is tempting to suppose that anything the family generates inherits the property one wants from a generated modal logic — *adequacy*, the characterisation of the theory’s behavioural equivalence by satisfaction. It does not, and the reason is worth stating before the construction rather than after it.

Adequacy is a property of the *target logic specification*, not of the generating machinery. The classical Hennessy–Milner argument needs a certain minimum of logical structure to run: something akin to a conjunction, so that a formula can record several observations of the same state at once, and something akin to a negation, so that a formula can record the *absence* of an observation. Strip either and the argument fails — without negation one characterises a simulation preorder rather than bisimulation; without conjunction one cannot separate states that differ only in how their capabilities are bundled. A generated logic lacking either connective will fail to have an adequacy

theorem, and no amount of care in the generator repairs this, because the deficiency is in what one asked the generator to produce.

Remark 8.1 (The Hypercube functor offers no adequacy theorem). The Hypercube algorithm generates a *type system*, and type systems are not generally expected to be adequate. Nobody asks of the simply-typed λ -calculus that its typing judgement characterise observational equivalence; one asks that it be sound, decidable, and useful. The same standard applies here. Sections 8.4 onwards should therefore be read as generating a classification discipline, not as generating a behavioural characterisation.

This is a deliberate trade, and Section 8.9 describes what is bought with it: the user dials which connectives are admitted, precisely in order to expose a surface on which a finitely checkable fragment is generable. The cost of that expressiveness and control is that not everything generated is necessarily adequate.

8.3 Input: rewrites and redex positions

The Hypercube algorithm takes as input a theory in the classifying form of Section 2.3. Its base rewrites have the schematic shape

$$x_1:X_1, \dots, x_n:X_n \mid \emptyset \vdash L(\vec{x}) \rightsquigarrow R(\vec{x}),$$

and — this is the crucial extra datum — it also tracks *redex positions*. Choosing a subterm occurrence t_j of L with carrier Y_j determines a one-hole context $K_j[-]$ with $K_j[t_j] = L$. Write

$$V_j = \text{FV}(K_j) - \text{FV}(t_j), \quad W_j = \text{FV}(t_j) - \text{FV}(K_j).$$

The variables in V_j — free in the context but not in the chosen subterm — become the *rely parameters* of the modality generated at that position.

Remark 8.2 (The relationship to the interaction cut). The reader will recognise K_j . In a theory presented as an interaction cut, the contact site K of $(\dagger)$ is a one-hole context in exactly this sense, and the distinguished positions are the two introductions. The interaction cut is thus the special case in which one privileges a particular family of redex positions, and the Hypercube algorithm is what one gets by not privileging them. This is why Section 3 does real work for the present section: it identifies which positions are canonical.

8.4 The modal layer

For each base rewrite and each chosen redex position, the algorithm introduces a primitive modality

$$\langle K_j \rangle_{\vec{x}::\vec{A}} B,$$

a type former at carrier Y_j , read as a *rely-possibly* specification: under the rely assumptions $\vec{x}::\vec{A}$, placing a term into the context $K_j[-]$ takes one step to a specified right-hand side inhabiting B .

$$\frac{(\Gamma \mid \Delta \vdash A_k :: s_k^{X_k})_{x_k \in V_j} \quad \Gamma, \vec{x}:\vec{X} \mid \Delta, \vec{x}::\vec{A} \vdash B :: s_{\text{out}}^{\text{Pr}}}{\Gamma \mid \Delta \vdash \langle K_j \rangle_{\vec{x}::\vec{A}} B :: s_{\text{out}}^{Y_j}} \text{ M-FORM}$$

$$\frac{\Gamma, \vec{w}:\vec{W}, \vec{x}:\vec{X} \mid \Delta, \vec{x}:\vec{A} \vdash B :: s_{\text{out}}^{\text{Pr}} \quad \Gamma, \vec{w}:\vec{W}, \vec{x}:\vec{X} \mid \Delta, \vec{x}:\vec{A} \vdash R(\vec{x}, \vec{w}) :: B}{\Gamma, \vec{w}:\vec{W} \mid \Delta \vdash t_j(\vec{w}) :: \langle K_j \rangle_{\vec{x}:\vec{A}} B} \text{M-INTRO}$$

Elimination is where a rewrite-generated modality differs from an equation-generated one. Because the modality comes from a *rewrite*, elimination does not provide full conversion; it provides the operational step to the specified right-hand side, and the typing of that right-hand side.

$$\frac{\Gamma \mid \Delta \vdash t :: \langle K_j \rangle_{\vec{x}:\vec{A}} B \quad (\Gamma \mid \Delta \vdash u_k :: A_k)_{x_k \in V_j}}{\Gamma \mid \Delta \vdash K_j[t][\vec{u}/\vec{x}] \rightsquigarrow R(\vec{u})} \text{M-STEP}$$

$$\frac{\Gamma \mid \Delta \vdash t :: \langle K_j \rangle_{\vec{x}:\vec{A}} B \quad (\Gamma \mid \Delta \vdash u_k :: A_k)_{x_k \in V_j}}{\Gamma \mid \Delta \vdash R(\vec{u}) :: B[\vec{u}/\vec{x}]} \text{M-ELIM}$$

To recover a conversion-like principle stable under rewriting, the algorithm freely adds an *empty-context possibility* operator $\Diamond$ at the carrier of programs, with $p :: \Diamond B$ asserting that p can take one step to some reduct of type B . Equation-generated modalities, by contrast, do support full conversion by the usual rule — they are “zero-step” steps.

8.5 The structural and propositional layers

The modal fragment is intentionally behavioural, and this has a consequence that should be stated plainly rather than discovered: *terms with no available one-step behaviour need not have any modal type at all*. A value, a normal form, a deadlocked process — none of these is classified by a layer whose formation rules are indexed by rewrites.

Two further layers repair this and extend expressivity.

Structural types. For each term former f the algorithm freely adds a type former $f^\sharp$ whose inhabitants are exactly those terms whose root former is f . These classify by the root of the parse tree, giving a baseline typing for programs which can then be combined with the modal layer to express shape and one-step behaviour in the same language. When the interaction constructor is read this way it becomes a separating-style connective: a term satisfies $K(\varphi_1, \varphi_2)$ when it splits across K into a part satisfying φ_1 and a part satisfying φ_2 . When K is associative-commutative this is a genuine separating conjunction in the sense of spatial and separation logics [4, 16]; when K is free it is a positional split.

Propositional operations, stratified. At each carrier and sort the algorithm adds propositional operations on type formers — but *stratified by the categorical strength required to interpret them*. The finite-limits fragment supplies $\top$ and $\wedge$. Additional structure is required for $\perp$, for finite joins and distributivity, and finally for implication and negation.

Remark 8.3 (What a specification must supply). The stratification is not a technicality of the metatheory; it is visible to the user of a generated logic. A specification whose classifying theory has finite limits and no more yields a conjunctive fragment; implication and negation become available only when the theory supplies the structure to interpret them. A developer writing a property is therefore working in a logic whose connectives are determined by their own specification, and a property that fails to elaborate may be failing for this reason rather than because it is false.

8.6 Sort slots and the equational centre

Why a *hypercube*? Each generated type former carries finitely many sort slots: one for each rely input and one for the output. Filling each slot with $*$ or $\square$ yields a raw Boolean cube of choices. But equational axioms and rewrite laws force certain slots to agree, and only the assignments stable under those laws survive. These form the *equational centre* $\mathcal{Z}$, and the vertices of the resulting hypercube are the elements of $\mathcal{Z}$.

The cube's axes are not postulated. They are determined locally by the free variables of operational contexts and the result types demanded by the rules.

8.7 Worked example: a rho-calculus modality

Fix arity 1. The communication rewrite is

$$n:\text{Nm}, p:\text{Pr}, \lambda x.q : [\text{Nm} \rightarrow \text{Pr}] \mid \emptyset \vdash \text{out}_1(n, p) \mid \text{in}_1(n, \lambda x.q) \rightsquigarrow q[@p/x].$$

Choose the continuation redex position

$$t_j := \lambda x.q, \quad K_j([-]) := \text{out}_1(n, p) \mid \text{in}_1(n, [-]),$$

so that $K_j[t_j]$ is the left-hand side. The free variables of K_j are $V_j = \{n, p\}$, hence the induced modality is $\langle K_j \rangle_{n::A_n, p::A_p} B$ at carrier $[\text{Nm} \rightarrow \text{Pr}]$, contributing three local sort slots

$$\{s_n^{\text{Nm}}, s_p^{\text{Pr}}, s_{\text{out}}\}.$$

Relative to the origin assignment $(*, *, *)$, every nonempty subset of the three slots can be lifted to $\square$, yielding $2^3 - 1 = 7$ local axes.

Now hold $s_n = *$ fixed. Operationally the channel n serves only to ensure that the input and output are co-addressed; the reduct $q[@p/x]$ does not otherwise depend on it. The remaining two-slot subgeometry is:

Vertex (at $s_n = *$)	s_p^{Pr}	s_{out}
simply-typed corner	$*$	$*$
polymorphism direction	$\square$	$*$
dependent-types direction	$*$	$\square$
type-constructors direction	$\square$	$\square$

This is Barendregt's square [3], recovered exactly. The three nontrivial directions correspond to the three extra edges of the Lambda Cube beyond the simply-typed corner. *The Cube is one face of one modality of one rewrite rule.* It arises here not because it was postulated but because the communication rule has two rely parameters and one output, and the sort slots of a two-parameter modality form a square.

Remark 8.4 (Why Barendregt needed no equational centre). In the Lambda Cube there are no built-in equations, and the underlying equations of β add no relations between sorts. The whole raw Boolean cube is therefore admissible, $\mathcal{Z}$ is everything, and the distinction between the raw cube and the centre never arises. It arises for us as soon as a specification carries equations — which, by rung two of the ladder, is essentially always.

8.8 The construction is a monad

Let $\mathbf{L}$ denote the category of theories in classifying form, with morphisms preserving Pr , $\rightsquigarrow$, and the internal-language structure, and let $\mathbf{L}_\Sigma$ denote the category of such theories equipped with the additional structure provided by the modal, structural, and propositional layers. On a morphism F , the action is by transport:

$$\langle K_j \rangle_{\vec{x}::\vec{A}} B \longmapsto \langle F(K_j) \rangle_{\vec{x}::F(\vec{A})} F \cdot B, \quad \Diamond B \longmapsto \Diamond(F \cdot B).$$

Proposition 8.5. *There is a forgetful functor $U : \mathbf{L}_\Sigma \rightarrow \mathbf{L}$ and a left adjoint $F : \mathbf{L} \rightarrow \mathbf{L}_\Sigma$ freely adjoining the layers, and the induced endofunctor*

$$\mathcal{K} = U \circ F : \mathbf{L} \longrightarrow \mathbf{L}$$

is a monad on $\mathbf{L}$: for each theory T , $\mathcal{K}(\mathsf{T})$ is the underlying theory of its free extension by generated type formers.

This completes the pattern of Part III. Three constructions, three monads, three free/forgetful resolutions:

$$\mathcal{C} \text{ (meter it)}, \quad \mathcal{H} \text{ (log it)}, \quad \mathcal{K} \text{ (type it)}.$$

In each case the left adjoint installs apparatus generated from the theory’s own presentation, and the right adjoint takes it away.

8.9 Three dials

Section 8.2 said that adequacy is a property of the target logic specification. This subsection is the other half of that statement: the target logic specification is something the *user* supplies, and supplying it is the design activity around which the whole apparatus is arranged.

Because the logic is generated from a language specification together with a choice of connectives, a user can select *fragments* to dial in finitely checkable properties. There are three axes:

- (1) **The language fragment.** Restrict the sub-signature over which properties are stated. A property quantifying over a decidable fragment of the terms is a different proposition from the same property over the whole theory.
- (2) **The connective fragment.** Restrict which of the generated connectives are in play, subject to Remark 8.3. The conjunctive–modal fragment behaves very differently from one closed under negation and implication.
- (3) **The cube vertex.** Choose an element of the equational centre $\mathcal{Z}$. This is the classical design choice — how much dependency to admit — now presented as a coordinate rather than a fork in the literature.

Each axis trades expressivity for checkability, and the three are independent. In the general case, where the user has chosen not to constrain things statically, an *inference token budget* prevents runaway property checks. The budget is the backstop, not the primary mechanism: the primary mechanism is fragment selection, and the budget catches what selection did not.

Remark 8.6 (Adequacy and checkability pull against each other). It would be convenient if the dials traded a property nobody wanted for one everybody did. They do not, and the tension is structural rather than incidental. Adequacy requires negation (Section 8.2); negation is among the

first connectives a user drops when chasing decidability, since it is what turns a search for a witness into a search over all possible witnesses. A user who dials down to a finitely checkable fragment has, in the typical case, dialled out of the adequate region on the way.

The right conclusion is not that one of the two properties is illusory but that they answer different questions. Adequacy asks whether the logic can *distinguish* exactly the programs the theory distinguishes — a question about the logic’s resolving power as a whole. Checkability asks whether a particular property can be decided of a particular program in bounded time. A user guarding a communication in Section 9.2 wants the second and has no use for the first. A user reasoning about when two implementations may be substituted wants the first and will accept a semi-decision procedure for it. The dials exist so that neither user has to pay for the other’s requirement.

Part IV — Implementation

9 Rholang 1.4: a DDL whose data can wiggle

9.1 Two sublanguages

Most languages have two sublanguages: a *data definition language* (DDL), in which one says what values there are, and a *control flow language* (CFL), in which one says what happens. Rholang 1.4 has both, and the interest of the arrangement lies in what each of them turns out to be.

The DDL is the **language!** block. It has a superpower: *the data can wiggle*. The reader has already seen why. A **language!** specification is exactly the triple of Definition 2.2, and the ladder of Section 2.4 is its expressivity story:

Blocks populated	What is presentable	Familiar name
types, terms	free term algebras	algebraic data types
+ equations	quotients of free algebras	finitely presentable universal algebra
+ rewrites	theories with dynamics	domain-specific languages

An ordinary DDL stops at the first row. Adding equations reaches the second, which is where a good many libraries secretly live — every time a developer documents that a data structure is “a list but order doesn’t matter” or “a string but compared case-insensitively,” they are hand-maintaining an equational presentation their type system cannot see. Adding rewrites reaches the third, and at that point the DDL is defining languages.

The CFL is merely a Rholang in which the channels are typed by **language!** specifications. That is the whole design. A channel’s type is a GSLT; sending on a channel means sending a term of that theory; and — because the theory comes with a generated logic by Section 8 — receiving on a channel can be guarded by a property expressed in that theory’s own logic.

9.2 The for-comprehension and its guard

The **for**-comprehension has grown a clause. Its generic form is

```

for( ptrn11 <- chan11 & ... & ptrnm1 <- chanm1 ;
    ... ;
    ptrn1n <- chan1n & ... & ptrnmn <- chanmn
    where condition ) P

```

where $\&$ joins within a group — simultaneity, all of the joined receives occurring together — and $;$ separates groups. The new clause is **where condition**, which gates the communication: the comm fires only if the condition holds. The preceding clauses bind variables, and those variables may be mentioned in the condition, so the guard is a condition on what was actually received.

The condition grammar is

$$\begin{aligned} \varphi, \psi &::= \text{true} \mid \text{false} \\ &\mid \text{ground formula from a language definition} \\ &\mid \sim\varphi \mid \varphi \& \psi \mid \varphi \setminus / \psi \mid \varphi \Rightarrow \psi \\ &\mid \text{spatial connectives from a language definition} \\ &\mid \langle K \rangle \varphi \end{aligned}$$

Every clause of that grammar has been constructed in Part III. The ground formulae and the spatial connectives are supplied by the language definition, which is to say by the structural layer of Section 8 — one ground formula per term former, and the interaction constructor read separating-style. The boolean connectives are the propositional layer, subject to the stratification of Remark 8.3. And $\langle K \rangle \varphi$ is the modal layer.

Remark 9.1 (One K , three altitudes). The K in $\langle K \rangle \varphi$ is not a new notion. It is the same object at three altitudes of the development. In Section 5 it labels transitions, and bisimulation — hence the identity of programs — is computed over context-labelled steps. In Section 8 it indexes the generated modality $\langle K_j \rangle_{\vec{x}::\vec{A}} B$, arising from a rewrite together with a choice of redex position. Here it is what a developer types in a **where** clause. The mathematical apparatus that makes the category work, the logic generated from a specification, and the surface syntax of a guard are three presentations of one thing.

Remark 9.2 (Budgets, and what a guard is for). Evaluating a guard is work performed at communication time, before the comm commits, and it is metered. Token budgets rein in infinitary computations and property checks alike; in the terms of Section 8.9, a user who has selected a finitely checkable fragment does not rely on the budget, and a user who has not is caught by it.

It is worth being explicit that the logic a guard is written in need not be adequate, and that this is not a compromise. A **where** clause asks whether *this* received term has *that* property, in bounded time. It does not ask the logic to resolve the theory’s behavioural equivalence. By Remark 8.6 the two demands pull in opposite directions, and the guard is squarely on the checkability side of that tension.

9.3 The rubric

The remainder of this section works one pattern four times. For each of four theories we

- (1) define the GSLT as a **language!** specification;
- (2) make an instance of a program in that GSLT and send it on a channel; and
- (3) receive the program on a channel and check it for a property.

The theories are the λ -calculus, Turing machines, the π -calculus, and the ambient calculus. They are chosen to span the discriminations of Part I: λ has a free K and a degenerate environment surface; Turing machines are a GSLT and nothing more; π migrates by binding; ambients migrate by spatial restructuring.

9.3.1 The λ -calculus

```
language! {
  name: Lambda,
  types { Term },
  terms {
    Lam . ^x.body:[Term -> Term] |- "lam " x "." body : Term;
    App . fun:Term, arg:Term      |- "(" fun "," arg ")" : Term;
  },
  equations { },
  rewrites {
    Beta      . |- (App (Lam fun) arg) ~> (eval fun arg);
    AppCongL . | M0 ~> M1 |- (App M0 N) ~> (App M1 N);
    AppCongR . | N0 ~> N1 |- (App M N0) ~> (App M N1);
    LamCong  . | S ~> T   |- (Lam ^x.S) ~> (Lam ^x.T);
  },
}
```

Send a term — here the self-application combinator applied to the identity:

```
new work in {
  work!( (lam x . (x , x) , lam y . y) )
}
```

Receive it and guard on a property. The property we want is “this term can take a β -step” — that is, the modality generated by the **Beta** rewrite at the top-level redex position:

```
for( t <- work where <Beta> true ) {
  // t is a Lambda term that has a beta-redex at the root
  reduce!( *t )
}
```

9.3.2 Turing machines

Section 3.4 argued that a Turing machine is a GSLT and not an interactive one. Both halves of that claim are visible here. The specification is unproblematic:

```
language! {
  name: Turing,

  types {
    Config
    Tape
    State
    Sym
  },

  terms {
    Blank . Sym ::= "_" ;
    Zero  . Sym ::= "0" ;
    One   . Sym ::= "1" ;

    Halt . State ::= "halt" ;
    Q     . n:UInt32 |- "q" n : State ;

    // tape as a zipper: reversed left context, head symbol, right context
    Tp . l:List(Sym), h:Sym, r:List(Sym) |- "<" l "|" h "|" r ">" : Tape ;
  },
}
```

```

    Cf      . q:State, t:Tape |- "(" q ", " t ")" : Config ;
  },

  equations { },

  rewrites {
    // one entry of the transition table, written out
    // q0 reading 0: write 1, move right, go to q1
    D_q0_0 . |- (Cf (Q Ou32) (Tp L Zero R))
              ~> (Cf (Q 1u32) (shift_right L One R));
    D_q0_1 . |- (Cf (Q Ou32) (Tp L One R))
              ~> (Cf Halt (Tp L One R));
  },
}

```

What one cannot write is an interaction constructor. There is no term former K taking a **Config** and an environment of the same sort into contact, because the only thing a configuration meets is its own tape — and **Tape** is not **Config**. The specification is accepted, the machine runs, and the theory sits at “GSLT only” in Table 2.

Sending and guarding nevertheless work, because they are operations of the CFL, not of the theory:

```

new work in {
  work!( (q0 , <[] | 0 | [0,1]>) )
  |
  for( c <- work where <D_q0_0> true ) {
    // c is a Config whose next step is the write-1-move-right transition
    step!(*c)
  }
}

```

This is the presentation/encoding distinction of Section 3.4 made concrete. The channel **work** is a Rholang channel; the interaction happens at Rholang’s cut, not at any cut of the Turing theory. What the guard inspects is a **Turing** term; what does the inspecting is a rho comm.

9.3.3 The π -calculus

```

language! {
  name: Pi,

  types { Proc Name },

  terms {
    PZero . Proc ::= "0" ;
    PNew . ^x.p:[Name -> Proc] |- "new" "(" x ", " p ")" : Proc ;
    PIn . n:Name, ^x.p:[Name -> Proc] |- n "?" x "." p : Proc ;
    POut . n:Name, m:Name, p:Proc |- n "!" m "." p : Proc ;
    PPar . ps:HashBag(Proc) |- "{" ps.*sep("|") "}" : Proc ;
    PRep . p:Proc |- "!" p : Proc ;
  },

  equations {
    NewComm . |- (PNew ^x.(PNew ^y.P)) = (PNew ^y.(PNew ^x.P)) ;
    ScopeExt . | x # ...rest
              |- (PPar {(PNew ^x.P), ...rest}) = (PNew ^x.(PPar {P, ...rest})) ;
    RepUnfold . |- (PRep P) = (PPar {P, (PRep P)}) ;
  }
}

```

```

},
rewrites {
  Comm . |- (PPar {(PIn n ^x.p), (POut n m q), ...rest})
    ~> (PPar {(subst ^x.p m), q, ...rest}) ;

  ParCong . | S ~> T |- (PPar {S, ...rest}) ~> (PPar {T, ...rest}) ;
  NewCong . | S ~> T |- (PNew ^x.S) ~> (PNew ^x.T) ;
},
}

```

The interaction-cut data is legible directly from the `Comm` rule: K is `PPar`, whose `HashBag` carrier makes it associative-commutative; the introductions are `PIn` and `POut`; the surfaces are the subjects n , matched by name equality; the continuations are $\lambda x.p$ and (m, q) ; and the contraction is pseudo-application, migrating m into p through the binder x .

Send a π -term and guard on a spatial property — here, that the received process splits into a part that can communicate and an arbitrary remainder:

```

new work in {
  work!( new(c, { c?y.0 | c!c.0 }) )
  |
  for( p <- work where PPar( <Comm> true , true ) ) {
    schedule!(*p)
  }
}

```

The connective `PPar(-, -)` is the structural type former generated from the term former `PPar`, read separating-style; because `PPar` is associative-commutative, it is a genuine separating conjunction.

9.3.4 The ambient calculus

```

language! {
  name: Ambient,
  types { Proc Name },
  terms {
    PZero . Proc ::= "0" ;
    PIn . Proc ::= "in(" Name ", " Proc ")" ;
    POut . Proc ::= "out(" Name ", " Proc ")" ;
    POpen . Proc ::= "open(" Name ", " Proc ")" ;
    PAmb . Proc ::= Name "[" Proc "]" ;
    PNew . ^x.p:[Name -> Proc] |- "new" "(" x " " p ")" : Proc ;
    PPar . Proc ::= HashBag(Proc) sep "|" delim "{" "}" ;
  },
  equations {
    NewComm . |- (PNew ^x.(PNew ^y.P)) = (PNew ^y.(PNew ^x.P)) ;
    ScopeExt . | x # ...rest
      |- (PPar {(PNew ^x.P), ...rest}) = (PNew ^x.(PPar {P, ...rest})) ;
    AmbNew . | x # P |- (PAmb N (PNew ^x.P)) = (PNew ^x.(PAmb N P)) ;
  },
  rewrites {
    InRule . |- (PPar {(PAmb N (PPar {(PIn M P), ...rest1})), (PAmb M R), ...rest2})
      ~> (PPar {(PAmb M (PPar {(PAmb N (PPar {P, ...rest1})), R))), ...rest2}) ;

    OutRule . |- (PAmb M (PPar {(PAmb N (PPar {(POut M P), ...rest1})), R, ...rest2}))
      ~> (PPar {(PAmb N (PPar {P, ...rest1})), (PAmb M R), ...rest2}) ;
  }
}

```

```

    OpenRule . |- (PPar {(POpen N P), (PAmb N Q), ...rest})
                ~> (PPar {P, Q, ...rest}) ;

    ParCong . | S ~> T |- (PPar {S, ...rest}) ~> (PPar {T, ...rest}) ;
    AmbCong . | S ~> T |- (PAmb N S) ~> (PAmb N T) ;
    NewCong . | S ~> T |- (PNew ~x.S) ~> (PNew ~x.T) ;
  },
}

```

This is the instance in which the contraction is neither a release nor a substitution. **OpenRule** dissolves the boundary of an ambient, releasing its contents into the enclosing context; **InRule** and **OutRule** relocate a subtree across a boundary. In every case **compute** is a spatial restructuring: no datum migrates through a binder, yet the configuration’s nesting — and hence what is near what — is rearranged. The ambient calculus is therefore the cleanest instance in which the contraction is a tree rewrite, and its presentation in the style of Cardelli and Gordon is what Section 2 was pointing at.

Send an ambient term and guard on its capacity to dissolve a boundary:

```

new work in {
  work!( { open(n, 0) | n[ m[0] ] } )
  |
  for( a <- work where <OpenRule> true ) {
    unpack!(*a)
  }
}

```

10 Conclusion

The through-line of this document is a single claim about where effort belongs. A language presented as (Σ, E, R) is not merely documented by its presentation; it is *determined* by it, in a strong enough sense that the artifacts one normally builds by hand become instances of constructions that do not know which language they are applied to.

Part I isolated the structure those constructions require, and isolated it in stages so that the requirements could be seen separately. A bare GSLT is a rewrite theory. An interactive GSLT additionally names a site — and the three machines of Section 3.4 are in the document precisely because they show that naming a site is a real condition, failed by theories no one would call impoverished. A continued interactive GSLT presents the site as a cut, supplies a canonical form, and tolerates decoration of its continuations.

Part II built the category, and the build was not routine: contexts, not terms, are the primary structure, because the canonical encodings are not maps of terms and because ambient contexts in a richer theory observe more than the encoded contexts of a poorer one.

Part III exhibited three constructions on that category, all of the same shape — $\mathcal{C}$ meters, $\mathcal{H}$ logs, $\mathcal{K}$ types. Each is a monad, each resolves through a free/forgetful adjunction, and each, over a sufficiently expressive base, can alternatively be dissolved into the base’s own computation.

Part IV closed the loop. The **language!** block is a data definition language whose three rungs are exactly the three components of Definition 2.2, and the control flow language is a Rholang whose channels carry those specifications as types. The **where** clause is where the parts meet: a guard written in the logic generated from the channel’s own type, over the modality $\langle K \rangle \varphi$ whose K is the same K that labels the transitions over which program identity is decided.

Open threads

Three, recorded for the reader who wants to push.

The stacking. $\mathcal{C}$ and $\mathcal{H}$ compose, and the composite has a reading: running $\mathcal{C}$ on an $\mathcal{H}$ -equipped model gives the cost of making the recording. The tower $\mathcal{CHCH}\cdots$ and its fixed point are objects we have not investigated.

Budget semantics. A guard whose evaluation exhausts its inference budget must do *something*, and the choices — evaluate to false, block, fault — are behaviourally distinct and therefore visible to the bisimulation of Section 5. Negation over a truncated check is where the choice bites hardest. This document specifies the budget and not the semantics of its exhaustion.

Self-hosting. The rho calculus is presentable as a **language!** specification; the CFL is a Rholang. The DDL can therefore define the CFL, and the strength of the resulting self-hosting claim — and what it would take to make it precise — is left open here.

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